Claudio Cellerini

Physics of Complex Systems Graduate

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Education

Academic

Mar 2022 - Apr 2025 **Master's Degree: Physics of Complex Systems**, *Università degli Studi di Torino*, Torino, Final Grade: 110/110 Cum Laude

Thesis: "Activity-Driven modeling of financial transactions networks"

- I developed and empirically validated a mechanistic, agent-based generative model for financial transactions networks. The model has been calibrated and validated on real transaction data

Key Academic Projects & Coursework

Laboratory of Multi-Agent Systems, *30/30*, URL

- Hands-on project: I led a three-person team in developing an agent-based financial market model, running simulations, benchmarking results against the literature, and analyzing discrepancies with real data
  GitHub Repository

Numerical Algorithms for Physics, *30/30 Cum Laude*, URL

- Hands-on project: I developed numerical methods in C++ to study chaotic pendulums, compared integration algorithms, and analyzed emergence of nonlinear dynamics
  Github Repository

Data Mining, URL

- Hands-on project: I developed and evaluated classifiers to predict the gender of Reddit users, leveraging both text content and subreddit participation
  Kaggle competition page, GitHub repository

Neural Networks, *30/30*, URL

Introduction to Games Theory and Network Theory, *30/30*, URL

Stochastic Processes for Physics, *30/30 Cum Laude*, URL

Complexity Economics, *Attended*, URL

Advanced Machine Learning, *Attended*, URL

Sep 2013 - Dec 2021 **Bachelor's Degree: Physics and Astrophysics**, *Università Degli Studi di Firenze*, Firenze, Final Grade: 96/110

Thesis: "Minority Game: a simplified market model"

- I carried on a comprehensive survey of the literature, and reproduced key result through simulation and data analysis

Self-directed

Apr 2025 - Present **Boot.dev – Software Development Bootcamp**, *Self-directed*, Curriculum URL

- Backend-developement training in Python, Go, SQL, Linux, Git

Professional Experience

Research Experience

Apr 2025 - * **Research Collaboration**, *External (CENTAI Institute)*, Torino

- I am contributing to research paper on financial transaction modeling as a co-author, and maintainer of the codebase

Mar 2024 - Apr 2025 **Master Thesis Research**, *CENTAI Institute*, Torino

- I worked in a research team on financial transaction network modeling, built experience in Python data analysis and model calibration/validation, with a solid project management through weekly updates and shared GitHub workflows

Teaching

Sep 2025 - * **Tutor**, *Il forno delle idee*, Torino

- I'm providing individualized tutoring for students with special educational needs

Nov 2024 **Teaching Assistant**, *King's College London (LISS DTP)*, London

- I supported a PhD-level seminar on Social Network Analysis, guiding students in using educational software and understanding its underlying concepts

Sep 2015 - Mar 2022 **Private Mathematics & Physics Tutor**, *Freelance*, Firenze

- I provided individualized tutoring in mathematics and physics to secondary school and university students, designing personalized study plans and supporting exam preparation

Sep 2015 - Mar 2020 **Museum Guide & Science Communicator**, *"Il Giardino di Archimede"*, Firenze

- I delivered interactive educational tours, communicating mathematics and physics concepts to audiences of varying ages and backgrounds

Commercial

May 2014 - Sep 2015 **Sales Representative**, *Master Agent*, Empoli

- I managed sales and support for utility contracts, delivering customized solutions for client

Technical Skills

Python Ecosystem

- Data analysis, manipulation, visualization: pandas, numpy, matplotlib [advanced]
- Statistical analysis, simulation, scientific computing: scipy, statsmodel, seaborn, pycop [intermediate]
- Machine learning: scikit-learn, tensorflow, pytorch [beginner]
- Paradigms: object-Oriented Programming [intermediate], functional programming [beginner]

Programming languages

- Compiled: C/C++ [intermediate], Mojo [novice]

Productivity packages and tools

- Spreadsheets: MS Excel, LO Calc, Google Sheets [intermediate]
- Presentation: MS PowePoint, Google Presentations [intermediate]
- Writing: Latex, Markdown, MS Word [intermediate]

Development tools

- Linux Bash shell [intermediate]
- SSH, Git[beginner]

Talks and Conferences

Jul 2025 **Regular Talk (accepted)**, *Complexity in Economics & Finance Workshop, StatPhys29 satellite (FinEcoNets & Econophysics Colloquium)*, Centro Ricerche "Enrico Fermi" (CREF), Rome, Italy

- Extended abstract *"Activity-Driven Modeling of Financial Transactions Networks"* accepted for a regular talk. *Talk not delivered due to non-attendance.*

Jul 2024 **Master Thesis Mid-term Seminar**, *University of Turin*

- Presented the progress of the master thesis project in a departmental seminar, discussing preliminary results and research directions.

Languages

Italian	Native
English	C1 - Fluent
Spanish	C1 - Fluent
French	A2 - Beginner